

```
theory Week_1_sol
imports Main
begin
```

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Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

lemma

```
assumes T: T b
and A: A a  $\wedge$  A b  $\implies$  a = b
and TA:  $\bigwedge x. T x \implies A x$ 
and Aab: A a
shows T a
```

proof—

— Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

```
have 1: A b
using TA[OF T]
.
```

— . proves the goal directly by a fact given with *using*, instantiating its variables if needed.

```
have 2:  $A\ a \wedge A\ b$   
  apply(rule conjI)  
  using Aab 1  
  .
```

```
have 3:  $a = b$   
  apply(rule A)  
  using 2  
  .
```

```
show ?thesis  
  using T  
  apply(subst 3)  
  .  
qed
```

```
end
```